PERFORMANCE OVERVIEW

Chiralight standard-motor

Native speed control.
No throttle-endpoint backoff.

ArduPilot: 95% endpoint

More of your battery is useful when the motor drive retains thrust at lower voltage. This simulated 3 kg quad uses 10-inch props, a 640KV motor and a 6S 4 Ah pack, while retaining a **3:1 thrust-to-weight ratio (TWR)** maneuver reserve for gust recovery and obstacle avoidance.

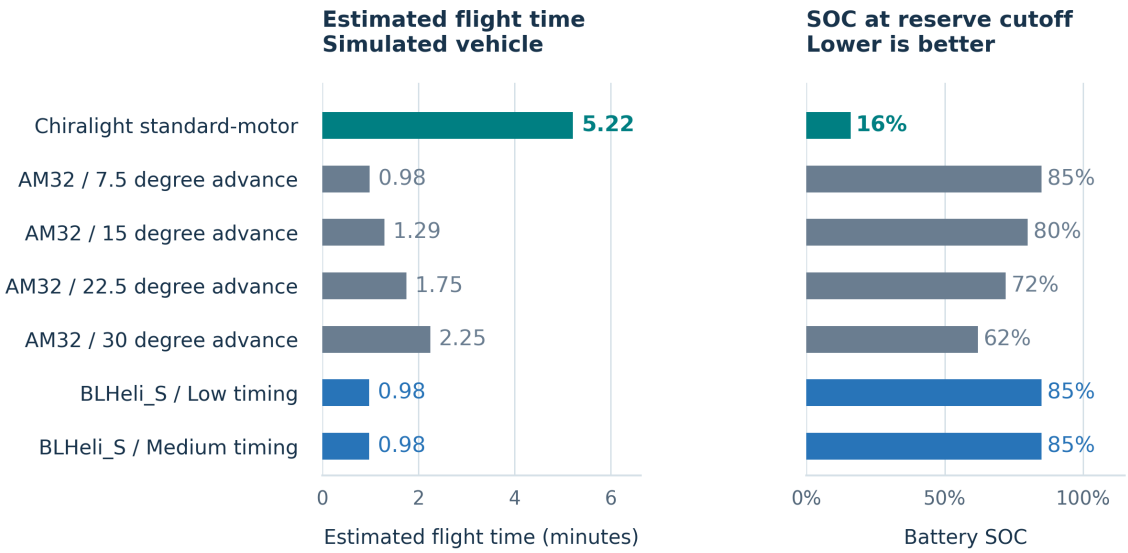
Up to 100%
or greater
More modeled flight time
Simulation results

Gain margin
More voltage headroom
for your chosen motor

+174 g
Potential payload by
choosing 6S over 8S

3:1 TWR MANEUVER RESERVE FOR GUST RECOVERY AND OBSTACLE AVOIDANCE

640KV / 6S | 3:1 TWR maneuver reserve

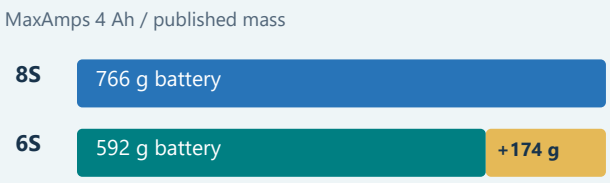


Estimated flight times for a simulated 3 kg vehicle at a 1.5:1 TWR flight load, with 3:1 TWR maneuver reserve.

95% throttle endpoint: tested 6S alternatives lose 3:1 TWR reserve at 89-99% SOC, or cannot reach it (page 6).

6S FOR PAYLOAD. 8S FOR ADDITIONAL VOLTAGE RESERVE.

The smaller 4 Ah pack releases **174 g for payload** at the same 3 kg all-up mass. The larger pack stores more energy and gives the drive more voltage headroom.



Similar energy, less power: MaxAmps 8S 2.8 Ah vs 6S 4 Ah (83 vs 89 Wh), but **38% lower peak and 27% lower continuous rated power**. Matching Wh does not guarantee sustained maneuver reserve.

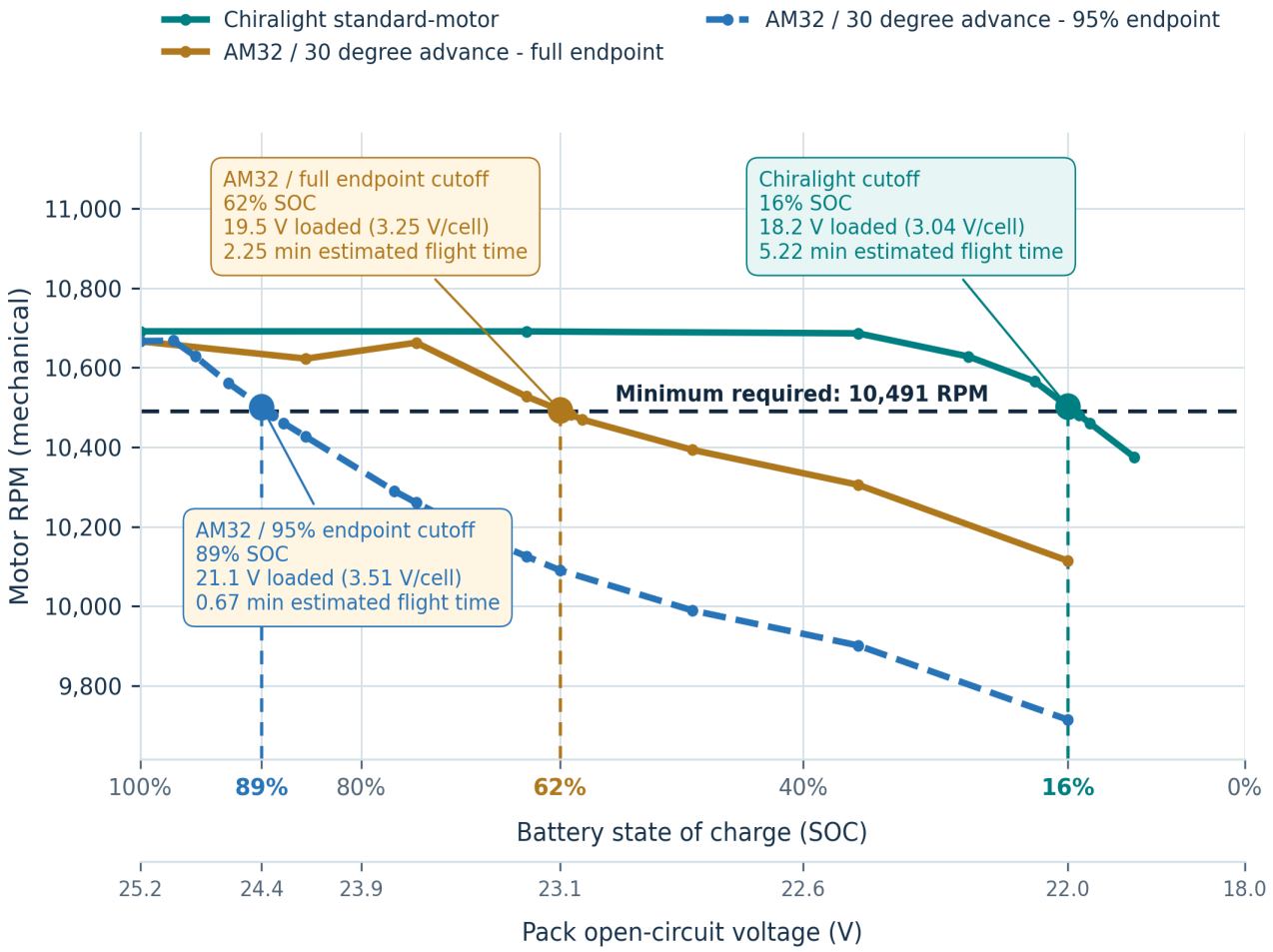
Reserve-cutoff rule: the selected SOC must retain the required thrust and at least 3.0 V/cell under the maneuver load. All ESCs receive adequate DC-link buffering; sustained battery resistance loss remains. Flight-time estimates use the simulated 1.5:1 TWR load; headline versus AM32 at 30 degree advance.

Verifiable simulation results. Contact Chiralight Systems for details.
Evaluate for your aircraft. Exact performance may vary.

BATTERY RESERVE

Motor speed and battery reserve

Chiralight standard-motor holds the required RPM deeper into battery discharge, using the available voltage range without a fixed throttle-endpoint reduction.



EXTRACT THE PERFORMANCE YOUR MOTOR AND BATTERY CAN DELIVER

Direct RPM regulation. Full modulation range. More usable battery.
Keep the 3:1 TWR maneuver reserve for gust recovery and obstacle avoidance, then use the voltage margin for longer missions or more flexible motor and battery sizing.

ArduPilot reference: default 95% throttle endpoint for thrust flattening near full throttle.

MOTOR AND DRIVE PERFORMANCE

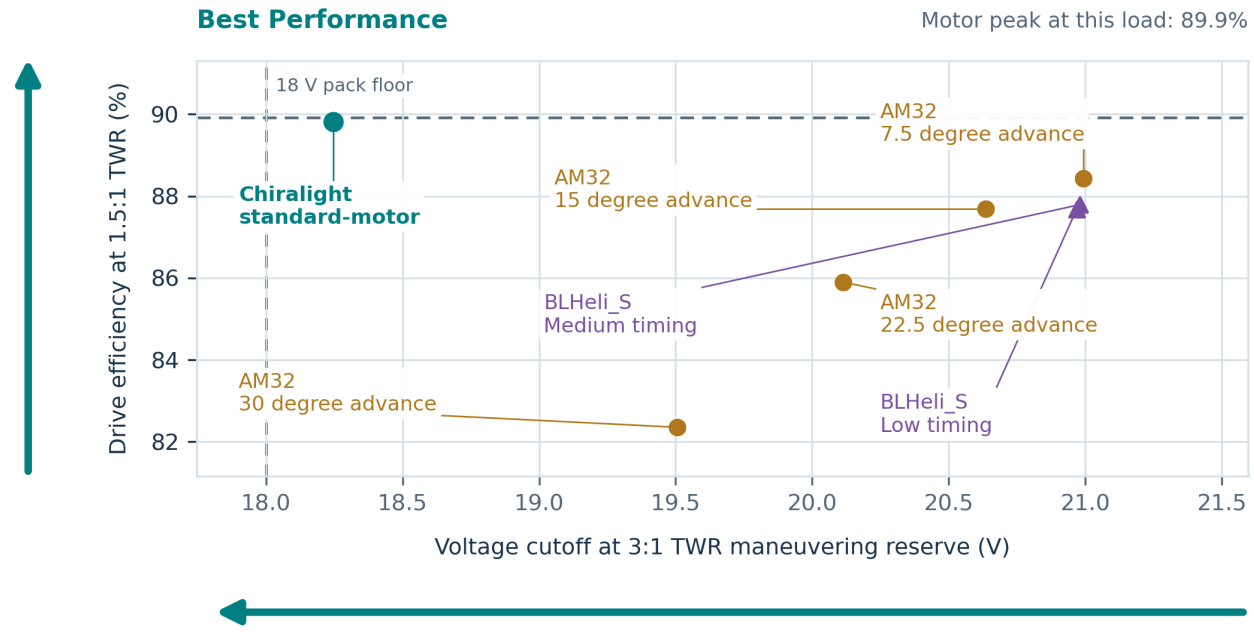
Efficiency and voltage headroom

Measured 640KV motor electrical parameters, nonlinear B/H and the same table-informed 10-inch propeller load. Actual compiled controllers; FOC field weakening off, MTPA and overmodulation on.

1.5:1 TWR FLIGHT EFFICIENCY | 3:1 TWR RESERVE CUTOFF

Controller / timing	Drive efficiency	Battery to shafts	3:1 TWR cutoff voltage
Motor theoretical peak (same load)	**89.9%**	--	--
Chiralight standard-motor FOC	**89.8%**	85.0%	18.2 V
AM32 / 7.5 degree advance	**88.4%**	83.7%	21.0 V
AM32 / 15 degree advance	**87.7%**	82.9%	20.6 V
AM32 / 22.5 degree advance	**85.9%**	81.2%	20.1 V
AM32 / 30 degree advance	**82.4%**	77.7%	19.5 V
BLHeli_S / Low timing	**87.7%**	82.9%	21.0 V
BLHeli_S / Medium timing	**87.8%**	82.9%	21.0 V

Flight efficiency: full-charge 6S at 1.5:1 TWR. Motor peak: ideal MTPA, copper loss only. Reserve: mean loaded voltage at the checked 3:1 TWR cutoff, buffered source and 3.0 V/cell floor. Pack: 4 Ah, 5 milliohms/cell.



The motor sets the efficiency ceiling. Chiralight standard-motor delivers 89.8% against this motor's 89.9% ideal ceiling at this flight load. Its voltage headroom gives you more freedom to choose motor size and KV for lower losses at your flight load, while preserving the required thrust reserve. **A more efficient motor choice translates directly into higher system efficiency.**

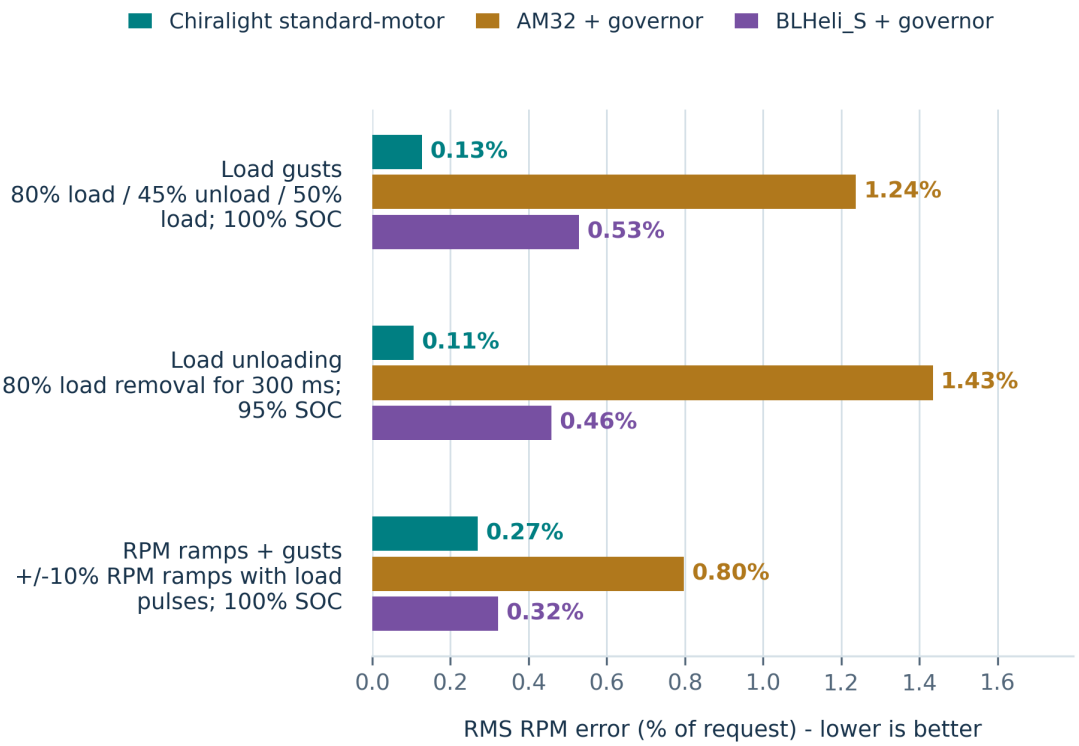
Verifiable simulation results. Contact Chiralight Systems for test details and supporting data.
Evaluate for your aircraft. Exact performance may vary.

CLOSED-LOOP PERFORMANCE

Dynamic response and startup

Same measured 640KV motor parameters, nonlinear B/H, table-informed propeller and 6S / 4 Ah battery at 5 milliohms/cell. Baseline: 1.5:1 TWR. Each controller uses its sensorless speed estimate; rotor truth only scores the result. Operating SOC is specified for each test.

MATCHED LOAD CHANGES; LOWER RPM ERROR IS BETTER



Separate speed-step training selected the loop settings: Chiralight standard-motor 80 rad/s; AM32 20 rad/s (integral gain x0.5); BLHeli_S 40 rad/s. Gains were frozen before these tests. AM32 and BLHeli_S use external governors driven by native commutation periods.

NATIVE STARTUP AND PROTECTION DIAGNOSTICS

Chiralight standard-motor

3/3 reached control. 3/3 completed; 0 phase trips.

AM32

3/3 reached control. 3/3 completed; 0 phase trips. 1 native startup recovery.

BLHeli_S

3/3 reached control. 3/3 completed; 0 phase trips.

The cards report three selected-gain cold starts per controller. A common 120 A phase guard supplies bench protection. The unloading comparison uses 95% SOC to accept regenerated energy. The separate full-SOC stress diagnostic records energy the held-full battery could not accept.

Verifiable simulation results. Contact Chiralight Systems for test details and supporting data.
Evaluate for your aircraft. Exact performance may vary.

HARDWARE-AWARE SIMULATION

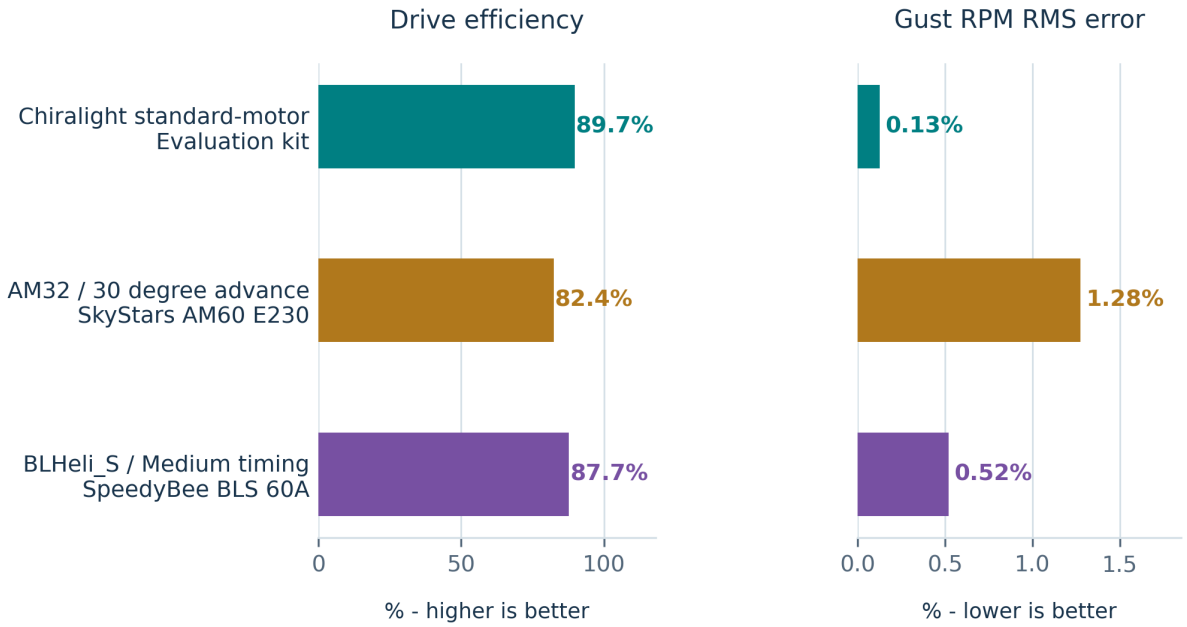
Inverter timing and current sensing

Chiralight evaluation-kit sensing and 104 ns blanking, alongside documented representative AM32 and BLHeli_S board profiles. The corrected motor, nonlinear B/H and propeller model remain the same.

SAME FLIGHT LOAD - BOARD-SPECIFIC ELECTRONIC BEHAVIOR

Controller / electronics	DC-to-shaft	Copper loss	Phase RMS
Chiralight standard-motor Chiralight evaluation kit	**89.7%**	21.5 W	12.0 A
AM32 / 30 degree advance SkyStars AM60 E230 profile	**82.4%**	38.3 W	16.0 A
BLHeli_S / Medium timing SpeedyBee BLS60A / J_H_40 profile	**87.7%**	25.6 W	13.1 A

Full-charge 6S / 4 Ah, 5 milliohms/cell, four motors and 10 W auxiliaries. Efficiency: settled 1.5:1 TWR flight load. Gust error: the frozen speed-loop settings from page 4. AM32 uses 30 degree advance, selected for voltage headroom.



TIMING AND MEASUREMENT PROFILES

Chiralight standard-motor: 104.0 ns blanking; 52.08 kHz carrier / 104.17 kHz API sampling; 13 ENOB across +/-250 A; 16-bit converter

AM32 / 30 degree advance: 416.7 ns blanking; 24.00 kHz carrier; native 12-bit DC bus-current ADC, 0-165 A

BLHeli_S / Medium timing: 816.3 ns blanking; 23.93 kHz carrier; no firmware current feedback

Modeled drive efficiency includes winding and diode losses. Unspecified device conduction, switching-energy and magnetic losses are omitted.

Verifiable simulation results. Contact Chiralight Systems for test details and supporting data.
Evaluate for your aircraft. Exact performance may vary.

TEST CONFIGURATION

Throttle endpoints and assumptions

At 6S, a 95% throttle endpoint makes the tested alternatives effectively unusable for this 3 kg vehicle's 3:1 TWR mission reserve. 2 of 6 settings miss the reserve even at full charge. The rest reach cutoff with 89-99% SOC remaining, allowing only about 4-40 seconds of modeled flight. Chiralight standard-motor retains 5.22 minutes.

3:1 TWR MANEUVER RESERVE - 6S / 4 AH, MEASURED 640KV MOTOR

Controller / timing	Full endpoint cutoff SOC	95% endpoint cutoff SOC	95% endpoint flight estimate
AM32 / 7.5 degree advance	85%	**Not met**	--
AM32 / 15 degree advance	80%	**Not met**	--
AM32 / 22.5 degree advance	72%	**96%**	0.26 min
AM32 / 30 degree advance	62%	**89%**	0.67 min
BLHeli_S / Low timing	85%	**99%**	0.07 min
BLHeli_S / Medium timing	85%	**98%**	0.13 min

Source: ArduPilot Motor Thrust Scaling, MOT_SPIN_MAX = 0.95 (default). This is a normalized ESC throttle endpoint, not a 5% deduction from RPM, thrust or efficiency. It is a configured comparison, not a stock-firmware restriction. 'Not met' means the reserve was not achieved even at full charge.

HOW TO READ THE PAGE 2 CHART

The chart plots minimum achieved mechanical RPM during the settled maneuver hold. The required 10,491 RPM corresponds to 2,250 gf per rotor and 3:1 TWR. Native operating points are joined without smoothing. The unrestricted AM32 bump is retained; it accompanies near-full-duty hunting in the tested AM32 and external RPM-governor configuration.

The voltage axis is pack open-circuit voltage from the SOC model. Callout voltages are the actual loaded DC link during the maneuver, including 5 milliohms/cell sag. Flight estimates use the separate 1.5:1 TWR flight load. The source assumes adequate DC-link buffering. Nonlinear B/H treatment, current limits and native controller timing are retained. The endpoint test changes the external command ceiling and its antiwindup bound; controller firmware, propeller and magnetic plant remain unchanged.

Verifiable simulation results. Contact Chiralight Systems for test details and supporting data.
Evaluate for your aircraft. Exact performance may vary.